

SACHIN KAROL

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Summary of Skills

Scientific editor with 13 years of experience and 2,400+ manuscripts reviewed across engineering, physical sciences, and life sciences. MS Bioengineering (University of Illinois at Chicago). Substantive editing, copyediting, and technical review for publication in *Nature*, *Science*, *Nano Letters*, and *Advanced Materials*. Also available for technical writing and documentation engagements.

Education

Ryerson University (now Toronto Metropolitan University) ◆ Toronto, Ontario, Canada

- Certificate in Data Analytics, Big Data, and Predictive Analytics (2018)

University of Illinois at Chicago ◆ Chicago, Illinois, USA

- Master of Science in Bioengineering (2011)

Bangalore University ◆ Bangalore, Karnataka, India

- Bachelor of Engineering in Instrumentation and Electronics (1999)

Recent Experience

Freelance Scientific Editor & Technical Writer, Self-employed

2026 – Present ◆ Toronto, Ontario

- Available for scientific editing and technical writing engagements
- Portfolio includes software user documentation and API reference samples

Cactus Communications, *Freelance Academic Editor, Physical Sciences and Engineering*

2011 – 2024 ◆ Canada/India

- Edited more than 2,400 manuscripts (over 6 million words) for publication in leading scientific journals such as *Nature*, *Science*, *Nano Letters*, and *Advanced Materials*
- Specialized in editing research articles related to the physical sciences and engineering disciplines with a focus on Materials Science
- Responsible for all aspects of the editing process, including proofreading, copyediting, and making substantive subject- and language-related changes to manuscripts
- Consistently delivered high-quality, error-free documents to the satisfaction of the client with 100% deadline conformance

University of Illinois at Chicago, Department of Restorative Dentistry, *Research Specialist, promoted from Research Assistant*

2007 – 2011 ◆ Chicago, Illinois

- Investigated the degradation of dental composite materials by oral bacterial biofilms using nanoindentation

- Determined the nanomechanical properties of resin-dentin bond components of different dental adhesive systems
- Analyzed both the short-term and long-term nanomechanical properties of dentin-resin-bonded interfaces treated with various chemical collagen cross-linkers
- Investigated the depth-dependent mechanical properties of nanostructured carbide-derived carbon using nanoindentation

Abstracts and Publications

- Karol, S., dos Santos, P.H., Bedran-Russo, A.K. Nanomechanical properties of intertubular dentin treated with collagen cross-linkers. American Association for Dental Research Symposium: Nanotechnology in the Health Sciences, Ann Arbor, Michigan, 2008.
- dos Santos, P.H., Karol, S., Bedran-Russo, A.K. Nanomechanical properties of biochemically modified dentin bonded interfaces. American Association for Dental Research Symposium: Nanotechnology in the Health Sciences, Ann Arbor, Michigan, 2008.
- Karol, S., Bedran-Russo, A.K. Nanomechanical properties of demineralized dentin treated with collagen-cross linkers. Academy of Dental Materials Meeting, Portland, Oregon, 2009.
- Karol, S., Bedran-Russo, A.K. Nanomechanical characterization of cross-linked demineralized carious dentin. IADR/AADR/CADR 89th General Session and Exhibition, San Diego, California, 2011.
- Castellan, C.S., Bedran-Russo, A.K., Karol, S., & Pereira, P.N. (2011). Long-term stability of dentin matrix following treatment with various natural collagen cross-linkers. *J Mech Behav Biomed Mater*, 4(7), 1343-1350.
- dos Santos, P.H., Karol, S., & Bedran-Russo, A.K. (2011). Long-term nanomechanical properties of biomodified dentin-resin interface components. *J Biomech*, 44(9), 1691-1694.
- dos Santos, P.H., Karol, S., & Bedran-Russo, A.K. (2011). Nanomechanical properties of biochemically modified dentin bonded interfaces. *J Oral Rehabil*, 38(7), 541-546.
- Ravindran, S., Gao, Q., Kotecha, M., Magin, R., Karol, S., Bedran-Russo, A., et al. (2011). Biomimetic ECM incorporated scaffold induces osteogenic gene expression in human marrow stromal cells. *Tissue Eng Part A*, 18(5-6), 489-501.
- Bedran-Russo, A.K., Karol, S., Pashley, D.H., & Viana, G. (2013). Site-specific properties of carious dentin matrices biomodified with collagen cross-linkers. *Am J Dent*, 26(5), 244-248.